MAD Step-Strip Lab

Unit 8: Sports Analytics Lab · Lesson 8-3 · EduWonderLab

UNIT 8 · LESSON 8-3 · TEACHER NOTES | **Standard 6.SP.5c**

FORMAT **Step-by-step strip lab**	TIME **25–35 min**
GROUPING **Pairs**	TOPIC **Mean Absolute Deviation**
STANDARD **6.SP.5c**	PREP LEVEL **Medium — print step strips and data cards**

Learning goal *I can find the mean absolute deviation (MAD) of a data set step by step.*

Teacher setup

- Print step strips and data cards per pair. Students arrange the steps in order.
- Steps: find the mean, find each distance from the mean, average the distances.
- Complete one data set, then check another group's steps.

Materials

Step strips, Data cards, Absolute deviation table, Pencil.

Differentiation & language support

- Distance-from-mean number line shows deviations (SPED).
- Sentence stem: "Each value is ___ away from the mean of ___."
- ESOL: use "distance from the mean" instead of "deviation" first.

Key vocabulary (English / Español):

Term	Español	What it means
mean absolute deviation	desviación media absoluta	the average distance from the mean
deviation	desviación	how far a value is from the mean

Quick teacher check: *Check the deviation sum on Card 2 (3 + 3 + 0 + 2 + 4 = 12).*

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Name: **Date:** **Class:**

Your goal: *I can find the mean absolute deviation (MAD) of a data set step by step.*

What you need

Step strips, Data cards, Deviation table.

How to play

1. Find the mean of the data set.
2. Find how far each value is from the mean (always positive).
3. Add the distances and divide by how many values there are.
4. Record the MAD.

Data Cards

Find the mean, then the distance of each value from the mean, then average those distances.

1. 2, 4, 6	**2.** 5, 5, 8, 10, 12
3. 10, 20, 30	**4.** 1, 3, 5, 7
5. 6, 8, 10, 12, 14	**6.** 3, 3, 3, 9

Deviation table (value, distance from mean):

Explain your thinking

Explain what a small MAD tells you about a data set compared to a large MAD.

Sentence frame: *Each value is about ___ away from the mean of ___.*

★ **Challenge:** *Cards 2 and 5 both have a MAD of 2.4. Explain how different data sets can have the same MAD.*

ANSWER KEY

Teacher copy — please do not distribute to students.

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Answers

1. 2, 4, 6 **Answer:** **mean 4; MAD = 4/3 ≈ 1.33**	**2.** 5, 5, 8, 10, 12 **Answer:** **mean 8; MAD = 12/5 = 2.4**
3. 10, 20, 30 **Answer:** **mean 20; MAD = 20/3 ≈ 6.67**	**4.** 1, 3, 5, 7 **Answer:** **mean 4; MAD = 8/4 = 2**
5. 6, 8, 10, 12, 14 **Answer:** **mean 10; MAD = 12/5 = 2.4**	**6.** 3, 3, 3, 9 **Answer:** **mean 4.5; MAD = 9/4 = 2.25**

Teaching notes & look-fors

- MAD = (sum of distances from the mean) ÷ (number of values).
- Card 6: mean 4.5; distances 1.5, 1.5, 1.5, 4.5 → 9 ÷ 4 = 2.25.